

CYBERSECURITY HOME LAB

SETUP RECORD - Phase 1: Kali Linux

Attacker VM



Prepared by	Abdullatif Abuzannad
Task	Complete Phase 1 (Kali Linux only) of the Cyber Lab Setup Guide v4.2
Hypervisor	Oracle VirtualBox (Windows host)
Guest OS	Kali Linux 2026.2 (pre-built VirtualBox image, amd64)
Date completed	10 August 2026

1. Objective

Build the attacker virtual machine for the practical pen-testing lab: a Kali Linux 2026.2 guest running under VirtualBox on an isolated NAT Network (10.0.0.0/24), configured with a fixed IP address and preserved with a snapshot. This covers Steps 1–6 of the guide. The victim machines (Windows 11/10/7, Android) are Phase 2 and are out of scope for this task.

2. Network Design

All lab machines sit on one custom NAT Network. Kali is given a fixed low address, so it is predictable that the victim machines will be added in Phase 2. DNS is resolved through the NAT gateway.

Virtual network type	NAT Network (custom)
Network name	NatNetwork
Subnet	10.0.0.0/24
DHCP	Enabled (per guide)
Kali address (static)	10.0.0.2 / 24
Gateway	10.0.0.1
DNS	10.0.0.1 (NAT gateway)

3. Step-by-Step Log

Step 1 — Install 7-Zip

- Downloaded from <https://7-zip.org/download.html> and installed with defaults.
- Required because the Kali VirtualBox image is distributed as a .7z archive.

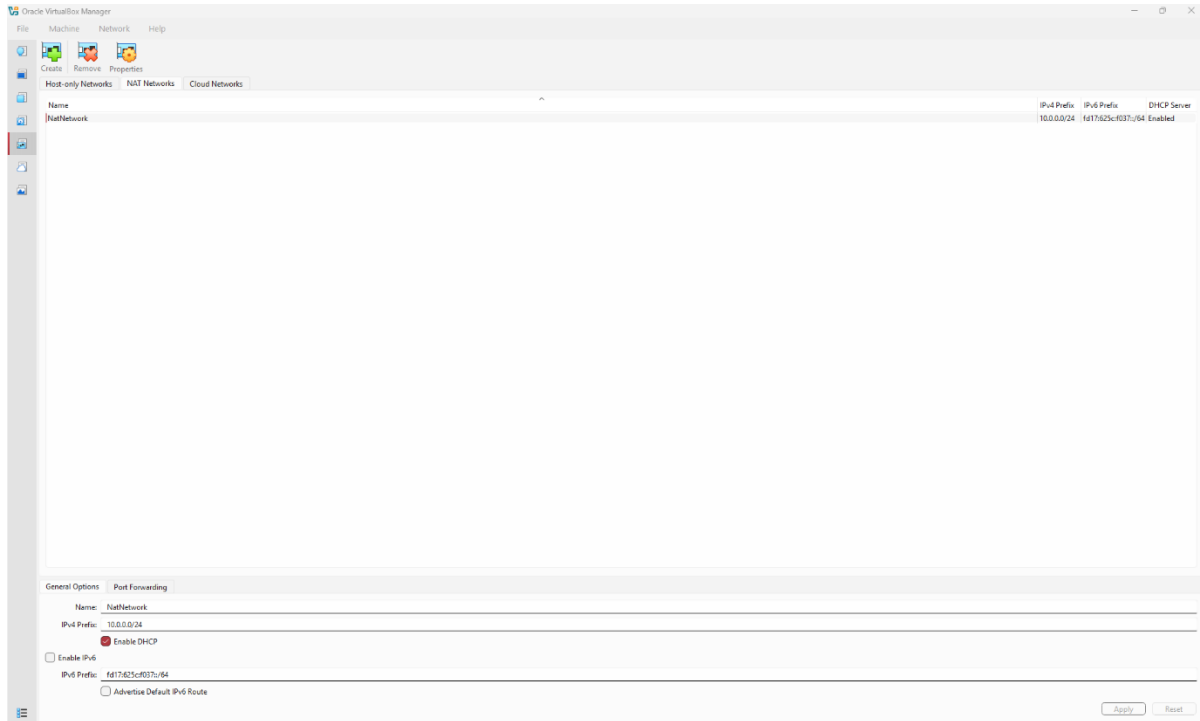
Step 2 — Install VirtualBox

- Downloaded the latest Windows-host build from <https://virtualbox.org/wiki/Downloads> and installed with defaults.



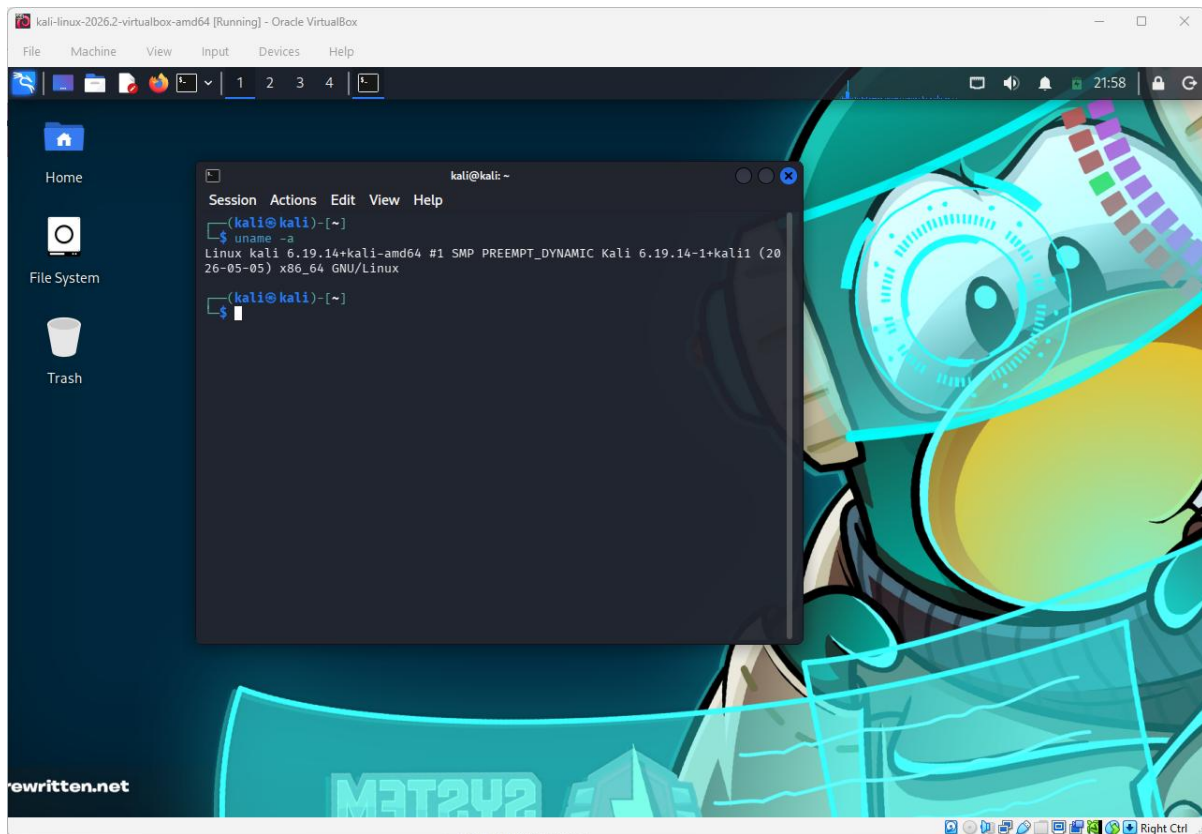
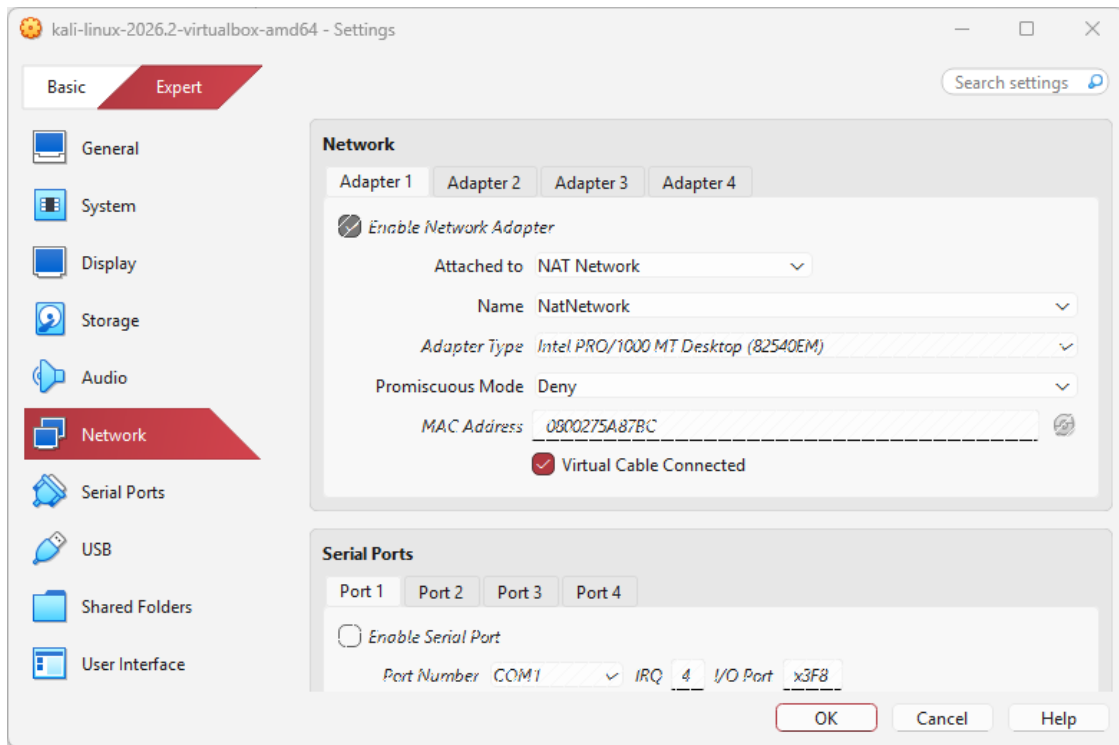
Step 3 — Create the NAT Network

- VirtualBox Manager → File → Tools → Network → NAT Networks → Create.
- Set Name = NatNetwork, IPv4 Prefix = 10.0.0.0/24, Enable DHCP = ticked, IPv6 = unticked → Apply.



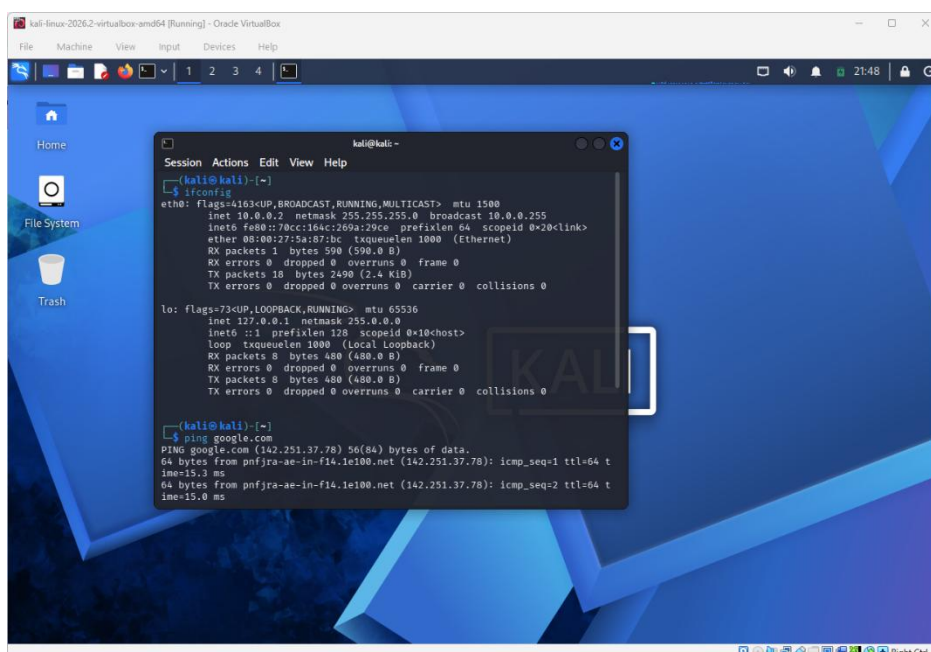
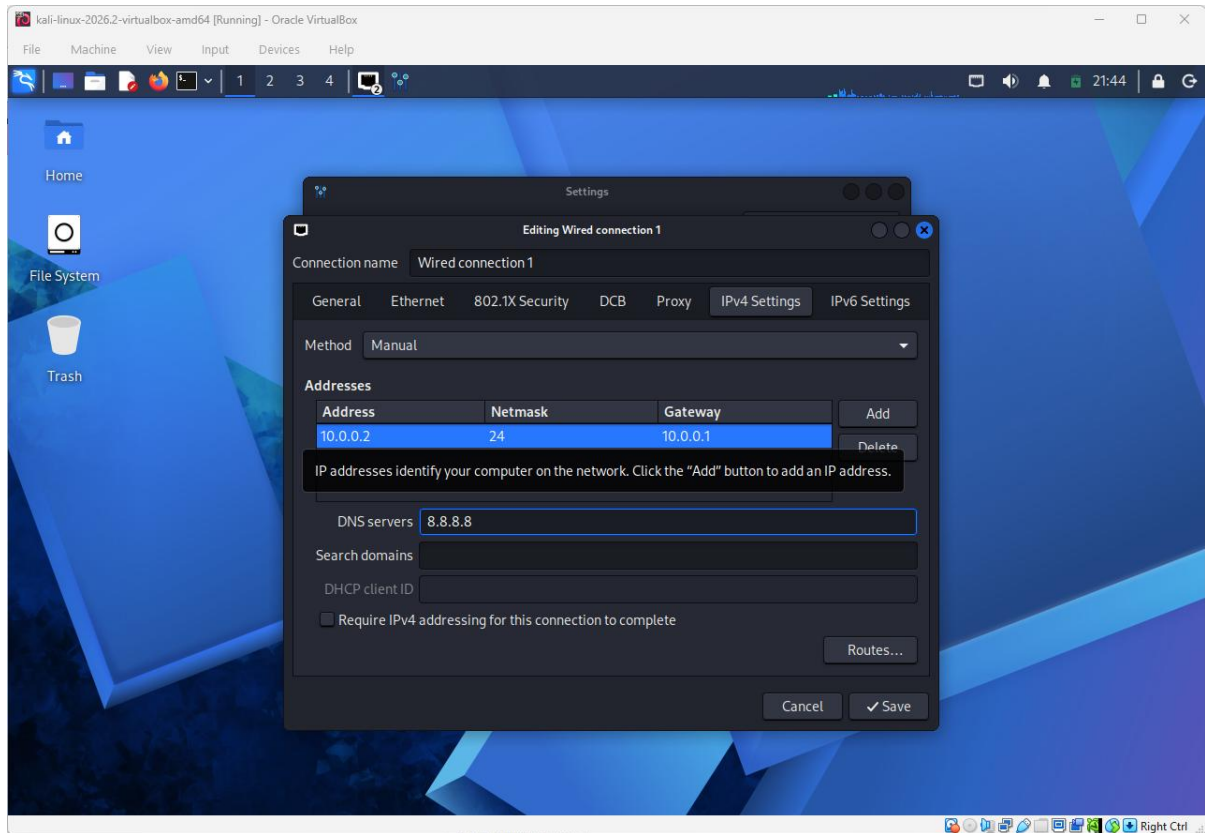
Step 4 — Download, verify, extract & import Kali

- Downloaded **kali-linux-2026.2-virtualbox-amd64.7z** from <https://kali.org/get-kali> (Virtual Machines → VirtualBox 64-bit).
- Verified that the SHA256 matched the value on the Kali download page.
- Applied resources (System: 4096 MB / 2 CPU; Display: 128 MB) and attached Adapter 1 to NAT Network “NatNetwork”.
- Booted, logged in with default **kali / kali**, and changed the password with **passwd**.



Step 5 — IP configuration of Kali

- Top-right network icon → Edit Connections → “Wired connection 1” → IPv4 Settings.
- Method = Manual → Address 10.0.0.2, Netmask 24, Gateway 10.0.0.1 → DNS servers 10.0.0.1 (NAT gateway) → Save. Toggled the connection off/on.



Step 6 — Take a snapshot (the Phase-1 deliverable)

- VirtualBox Manager → select the Kali VM → Snapshots → Take (VM powered off).
- Named **“Kali 2026.2 — Phase 1 baseline (static 10.0.0.2)”**, with a description recording the completed configuration.
- Serves as a clean restore point and the baseline for Phase 2.

